THE INFINITESIMAL LIFTING CRITERION FOR FORMALLY SMOOTH MORPHISMS OF ALGEBRAIC SPACES

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 4836-4857 of sources/stacks-spaces-more-morphisms/pinned-original-source.tex; proof span is lines 4859-4935, the complete author proof (75 body lines).

Editorial note. Navigation stubs: 16 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. Formally smooth morphisms, the small etale site, and the cohomological obstruction to smoothness; the implication is retained rather than cited.

Source reference. Original lemma definition-formally-smooth: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 4491–4496. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma lemma-action-sheaf: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 3827–3838. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma lemma-composition-formally-smooth-etale-unramified: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 2486–2495. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma lemma-etale-on-top: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 2567–2576. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma lemma-formally-smooth-sheaf-differentials: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 4769–4773. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Date: stacks-project:spaces-more-morphisms.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Source reference. Original lemma lemma-h1-is-zero: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 4803–4812. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma lemma-helper-formally-smooth: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 4554–4565. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma lemma-representable-property-formally-property: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 2529–2537. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma lemma-sheaf: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 3791–3805. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma lemma-smooth-formally-smooth: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 4587–4595. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma more-morphisms-definition-formally-unramified: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 4935–4935. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma remark-special-case: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 3848–3864. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma sites-cohomology-definition-torsor: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 4935–4935. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma sites-cohomology-lemma-torsors-h1: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 4935–4935. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma spaces-descent-lemma-locally-projective-descends: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 4935–4935. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma spaces-properties-lemma-locally-projective-pullback: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 4935–4935. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let S be a scheme. Let $f : X \to Y$ be a morphism of algebraic spaces over S. The following are equivalent:*

(1) *f is formally smooth,*
(2) *for every diagram*

$$\begin{array}{ccc} U & \xrightarrow{\psi} & V \\ \downarrow & & \downarrow \\ X & \xrightarrow{f} & Y \end{array}$$

where U and V are schemes and the vertical arrows are étale the morphism of schemes ψ is formally smooth (as in More on Morphisms, Definition more-morphisms-definition-formally-unramified), and
(3) *for one such diagram with surjective vertical arrows the morphism ψ is formally smooth.*

Proof. We have seen that (1) implies (2) and (3) in Lemma lemma-helper-formally-smooth. Assume (3). The proof that f is formally smooth is entirely similar to the proof of (1) $\Rightarrow$ (2) of Lemma lemma-smooth-formally-smooth.

Consider a solid commutative diagram

$$\begin{array}{ccc} X & \xleftarrow{a} & T \\ {\scriptstyle f}\downarrow & \nwarrow & \downarrow{\scriptstyle i} \\ Y & \xleftarrow{} & T' \end{array}$$

as in Definition definition-formally-smooth. We will show the dotted arrow exists thereby proving that f is formally smooth. Let $\mathcal{F}$ be the sheaf of sets on $(T')_{spaces,étale}$ of Lemma lemma-sheaf as in the special case discussed in Remark remark-special-case. Let

$$\mathcal{H} = \mathcal{H}om_{\mathcal{O}_T}(a^*\Omega_{X/Y}, \mathcal{C}_{T/T'})$$

be the sheaf of $\mathcal{O}_T$-modules on $T_{spaces,étale}$ with action $\mathcal{H} \times \mathcal{F} \to \mathcal{F}$ as in Lemma lemma-action-sheaf. The action $\mathcal{H} \times \mathcal{F} \to \mathcal{F}$ turns $\mathcal{F}$ into a pseudo $\mathcal{H}$-torsor, see Cohomology on Sites, Definition sites-cohomology-definition-torsor. Our goal is to show that $\mathcal{F}$ is a trivial $\mathcal{H}$-torsor. There are two steps: (I) To show that $\mathcal{F}$ is a torsor we have to show that $\mathcal{F}$ has étale locally a section. (II) To show that $\mathcal{F}$ is the trivial torsor it suffices to show that $H^1(T_{étale}, \mathcal{H}) = 0$, see Cohomology on Sites, Lemma sites-cohomology-lemma-torsors-h1.

First we prove (I). To see this consider a diagram (which exists because we are assuming (3))

$$\begin{array}{ccc} U & \xrightarrow{\psi} & V \\ \downarrow & & \downarrow \\ X & \xrightarrow{f} & Y \end{array}$$

where U and V are schemes, the vertical arrows are étale and surjective, and ψ is formally smooth. By Lemma lemma-representable-property-formally-property the morphism $V \to Y$ is formally étale. Thus by Lemma lemma-composition-formally-smooth-etale-unramified the composition $U \to Y$ is formally smooth. Then (I) follows from Lemma lemma-etale-on-top part (4).

Finally we prove (II). By Lemma lemma-formally-smooth-sheaf-differentials we see that $\Omega_{U/V}$ locally projective. Hence $\Omega_{X/Y}$ is locally projective, see Descent on Spaces, Lemma spaces-descent-lemma-locally-projective-descends. Hence $a^*\Omega_{X/Y}$

is locally projective, see Properties of Spaces, Lemma spaces-properties-lemma-locally-projective-pullback. Hence

$$H^1(T_{étale}, \mathcal{H}) = H^1(T_{étale}, \mathcal{H}om_{\mathcal{O}_T}(a^*\Omega_{X/Y}, \mathcal{C}_{T/T'}) = 0$$

by Lemma lemma-h1-is-zero as desired. □